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//SIT 55 : ASSIGNMENT 3 : CIRCULAR QUEUE
/*Implement Circular Queue using Array. Perform following operations on
it.
a) Insertion (Enqueue)
b) Deletion (Dequeue)
c) Display
(Note: Handle queue full condition by considering a fixed size of a
queue.)*/

#include <iostream>
using namespace std;

#define SIZE 20

class QUEUE {
private:
    int Que[SIZE];
    int front;
    int rear;

public:
    QUEUE(void) {
        front = rear = -1; // Initialize front and rear to -1
    }

    void addq(int item) {
        if ((front == 0 && rear == SIZE - 1) || (rear == (front - 1) %
(SIZE - 1))) {
            cout << "Queue is full." << endl;
            return;
        } else if (front == -1) {
            front = rear = 0;
            Que[rear] = item;
        } else {
            rear = (rear + 1) % SIZE;
            Que[rear] = item;
        }
    }

    int delq(void) {
        if (front == -1) {
            cout << "Queue is empty." << endl;
            return -1;
        }
        int item = Que[front];
        if (front == rear) {
            front = rear = -1;
        } else {
            front = (front + 1) % SIZE;
        }
        return item;
    }

    int qempty(void) {
        return (front == -1) ? 1 : 0;
    }

    int qfull(void) {
        return ((rear + 1) % SIZE == front) ? 1 : 0;
    }
}

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int frontale(void) {
    if (front == -1) {
        cout << "Queue is empty." << endl;
        return -1;
    }
    return Que[front];
}

int rearele(void) {
    if (front == -1) {
        cout << "Queue is empty." << endl;
        return -1;
    }
    return Que[rear];
}

void displayQ(void) {
    if (front == -1) {
        cout << "Queue is empty." << endl;
        return;
    }
    int i = front;
    do {
        cout << Que[i] << " ";
        i = (i + 1) % SIZE;
    } while (i != (rear + 1) % SIZE);
    cout << endl;
}

};

int main() {
    QUEUE Q;
    int ch, item, ans;
    int r; // Declare r outside the switch statement
    do {
        cout << "1. Add element:" << endl;
        cout << "2. Delete element:" << endl;
        cout << "3. Display element:" << endl;
        cout << "4. Display Front element:" << endl;
        cout << "5. Display Rear element:" << endl;
        cout << "6. Exit:" << endl;
        cout << "Enter your choice:" << endl;
        cin >> ch;
        switch (ch) {
            case 1:
                r = Q.qfull();
                if (r == 0) {
                    cout << "Enter the number to insert in the queue:" <<
endl;

                    cin >> item;
                    Q.addq(item);
                } else {
                    cout << "Queue is full." << endl;
                }
                break;
            case 2:
                r = Q.qempty();
                if (r == 0) {
                    ans = Q.delq();

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        cout << "Deleted element is: " << ans << endl;
    } else {
        cout << "Queue is empty." << endl;
    }
    break;
case 3:
    r = Q.qempty();
    if (r == 0) {
        Q.displayQ();
    } else {
        cout << "Queue is empty." << endl;
    }
    break;
case 4:
    r = Q.qempty();
    if (r == 0) {
        ans = Q.frontele();
        cout << "Front element is: " << ans << endl;
    } else {
        cout << "Queue is empty." << endl;
    }
    break;
case 5:
    r = Q.qempty();
    if (r == 0) {
        ans = Q.rearele();
        cout << "Rear element is: " << ans << endl;
    } else {
        cout << "Queue is empty." << endl;
    }
    break;
case 6:
    cout << "Thank you!!!!!!!" << endl;
    break;
    }
} while (ch != 6);

return 0;
}

```

/*OUTPUT:

```

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:
Enter your choice:
1
Enter the number to insert in the queue:
6
1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:
Enter your choice:
1

```

Enter the number to insert in the queue:

22

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

1

Enter the number to insert in the queue:

2

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

1

Enter the number to insert in the queue:

55

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

1

Enter the number to insert in the queue:

96

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

3

6 22 2 55 96

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

2

Deleted element is: 6

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

2

Deleted element is: 22

1. Add element:

2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

3

2 55 96

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

4

Front element is: 2

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

5

Rear element is: 96

1. Add element:
2. Delete element:
3. Display element:
4. Display Front element:
5. Display Rear element:
6. Exit:

Enter your choice:

6

Thank you!!!!!!*/